Logical Clifford Synthesis for Subsystem Codes: Counting Symplectic Solutions via Gauge Orbits

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***Abstract*—The Logical Clifford Synthesis (LCS) framework of Rengaswamy et al. (IEEE TQE, 2020) enumerates all physical Clifford circuits realizing a given logical Clifford operator for an $[[m,k]]$ stabilizer code, proving that exactly $2^{r(r+1)/2}$ symplectic solutions exist, where $r = m - k$. This result, however, does not extend to subsystem stabilizer codes, whose gauge group introduces degrees of freedom invisible to stabilizer-only theory. We formulate and solve the LCS problem for $[[m,k,g,d]]$ subsystem codes. Our main result proves that the group H_{gauge} of physical symplectic matrices normalizing the gauge group and acting trivially on logical qubits splits as a direct product $H_{\mathrm{gauge}} \cong H_{\mathrm{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2)$, where H_{stab} is the stabilizer-sector group of order $2^{r(r+1)/2}$ with $r = m - k - g$. Consequently every logical Clifford has exactly $2^{r(r+1)/2} \cdot |\mathrm{Sp}(2g, \mathbb{F}_2)|$ distinct physical realizations, carrying a product structure that permits independent optimization over stabilizer and gauge degrees of freedom. We supply a complete proof via a splitting of a short exact sequence, an efficient two-phase enumeration algorithm with $O(m^3)$ preprocessing and $O(m^2)$ per solution, and a verification on the $[[4,1,1,2]]$ Bacon-Shor code recovering all 48 distinct solutions versus the 8 predicted by the naïve stabilizer count. We further derive closed-form orbit-size formulas, discuss conditions under which the splitting may fail for certain non-Abelian gauge groups, and discuss implications for noise-adaptive compilation on near-term hardware.**

***Index Terms*—Quantum error correction, subsystem codes, Clifford synthesis, symplectic geometry, gauge freedom, fault-tolerant computation, Bacon-Shor codes, circuit compilation.**

I. Introduction

A. Motivation

Fault-tolerant quantum computation requires translating *logical* operators—unitaries acting on encoded qubits—into *physical* circuits on hardware qubits. This translation is never unique: for any logical operator there is a large family of physically inequivalent circuits all implementing the same logical action, and the choice has significant consequences for hardware performance. On near-term devices, qubit fidelities and coupling-link reliabilities vary widely and change with time [1], [2], so the optimal physical implementation of a given logical gate may differ from run to run. A systematic method for enumerating *all* valid physical implementations, and selecting the best among them for a given noise profile, is therefore of substantial practical importance.

For *stabilizer* codes, this problem was placed on a rigorous algebraic footing by Rengaswamy, Calderbank, Kadhe, and Pfister [3], whose Logical Clifford Synthesis (LCS) framework reduces circuit synthesis to linear algebra over the binary field $\mathbb{F}_2$. Their central result (Theorem 1 below) proves that for an $[[m,k]]$ stabilizer code with $r = m - k$ stabilizer qubits, every logical Clifford has *exactly* $|H_{\mathrm{stab}}| = 2^{r(r+1)/2}$ distinct physical symplectic realizations, and supplies a polynomial-time algorithm to enumerate them.

B. The Subsystem Gap

Subsystem stabilizer codes [4], [5] encode k logical qubits and g *gauge* qubits into m physical qubits, with parameters $[[m,k,g,d]]$. The gauge qubits carry no logical information, but elements of the gauge group $\mathcal{G}$ that lie outside the stabilizer group $\mathcal{S} = Z(\mathcal{G})$ generate additional freedom for physical circuit design: a physical Clifford need only *normalize* $\mathcal{G}$ (not fix $\mathcal{S}$ pointwise) to implement a valid logical operation. This broader class of valid implementations is entirely invisible to the original LCS theory, which imposes the stronger condition that the stabilizer be preserved element-wise.

More precisely, the stabilizer-only count of $2^{r'(r'+1)/2}$ with $r' = m - k$ treats all $m - k$ syndrome bits as stabilizers and applies to the quotient code obtained by gauge-fixing; it is therefore *inapplicable* to the unfixed subsystem code because the gauge space $W_{\mathcal{G}}$ is not isotropic—its non-degenerate part V_{gauge} has $\langle \mathbf{v}, \mathbf{v}' \rangle_s \neq 0$ for some $\mathbf{v}, \mathbf{v}' \in V_{\mathrm{gauge}}$, violating the isotropicity assumption of the LCS theorem.

Subsystem codes have attracted renewed interest for two reasons: their favorable syndrome-extraction properties (measurements can be performed on low-weight gauge operators rather than high-weight stabilizers), and their central role in modern constructions. The Bacon-Shor codes [5] demonstrated single-shot error correction [6]. The honeycomb Floquet code [7] and subsystem hypergraph product codes [8] are leading candidates for near-term hardware. A systematic LCS theory for these codes is therefore both timely and practically important.

C. Contributions

This paper makes the following key contributions:

1) **Problem formulation.** We define the subsystem LCS problem precisely (Section III), identifying the three conditions **(G)**, **(S)**, **(L)** that characterize valid physical realizations for subsystem codes, and explain why the original LCS conditions are strictly more restrictive.
2) **Structural decomposition.** We prove (Lemma 3) that the gauge space $W_{\mathcal{G}}$ decomposes orthogonally in the

symplectic sense as $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\text{gauge}}$, where V_{gauge} is a non-degenerate $2g$-dimensional symplectic subspace. This decomposition is the key algebraic input to the main theorem.
3) **Main theorem and proof.** Theorem 4 proves $H_{\text{gauge}} \cong H_{\text{stab}} \times \text{Sp}(2g, \mathbb{F}_2)$ via a split short exact sequence, yielding the closed-form count $|\text{Sol}(U_L)| = 2^{r(r+1)/2} \cdot |\text{Sp}(2g, \mathbb{F}_2)|$ for every logical Clifford U_L.
4) **Efficient algorithm.** Algorithm 1 enumerates all solutions in two phases—stabilizer solutions via the original LCS algorithm, and gauge orbits via enumeration of $\text{Sp}(2g, \mathbb{F}_2)$—with overall complexity $O(m^3)$ preprocessing plus $O(m^2)$ per solution.

Section V verifies the theory on the $[[4, 1, 1, 2]]$ Bacon-Shor code, recovering all 48 solutions and demonstrating that gauge-orbit solutions enable hardware-efficient implementations impossible in the stabilizer-only framework.

D. Related Work

Hardware-tailored LCS for stabilizer codes was formulated as an integer program in [9]. Logical Clifford gates via code automorphisms were studied in [10] for CSS and non-CSS stabilizer codes, and targeted symplectic constructions for hypergraph product codes appear in [11]. Logical Trotter circuits via the transvection correspondence were developed in [12]. None of these works addresses subsystem codes or derives the gauge-orbit product structure of our result.

The algebraic structure of subsystem codes has been studied through the lens of operator quantum error correction [4], [13] and through the theory of gauge-fixed codes [8]. Physical implementations of logical gates for specific subsystem codes (notably Bacon-Shor and Floquet codes) have been worked out case by case [7], [14], but no general enumeration theory existed prior to this work. A complementary code-construction perspective appears in [15], which derives algorithms for splitting stabilizer generators into lower-weight gauge operators while preserving logical Paulis; our work takes the resulting subsystem code as input and enumerates its full implementation space.

Extensions of LCS to non-Clifford operators in the third level of the Clifford hierarchy remain open [3]. Recent work on optimal Clifford synthesis [16], [17] provides efficient decomposition algorithms applicable to the output of our enumeration procedure. Noise-adaptive compilation for stabilizer codes was studied in [1], [2], [9]; our work provides the analogous foundation for subsystem codes.

E. Paper Organization

Section II reviews the symplectic representation of Paulis and Cliffords, recalls the original LCS theorem, and defines subsystem stabilizer codes. Section III formulates the subsystem LCS problem and proves the main decomposition theorem. Section IV presents and analyzes SubLCS, a two-phase enumeration procedure for all physical realizations of a given logical Clifford on a subsystem code, given as Algorithm 1. Section V works through the $[[4, 1, 1, 2]]$ Bacon-Shor code in detail. Section VI discusses compilation implications, limitations, and open problems.

TABLE I
KEY NOTATION USED THROUGHOUT THE PAPER. ALL VECTOR SPACES ARE OVER $\mathbb{F}_2$ AND EQUIPPED WITH THE SYMPLECTIC INNER PRODUCT $\langle \cdot, \cdot \rangle_s$.

Symbol	Meaning	Dimension/Order
m	Number of physical qubits	—
k	Number of logical qubits	—
g	Number of gauge qubits	—
r	Stabilizer rank: $r = m - k - g$	—
d	Code distance	—
$\mathbb{F}_2^{2m}$	Binary symplectic vector space for m qubits	$2m$
$\mathcal{W}_S$ (W_S)	Stabilizer subspace (isotropic)	r
$\mathcal{W}_G$ (W_G)	Gauge group subspace	$r + 2g$
$\mathcal{V}_{\text{gauge}}$ (V_{gauge})	Non-degenerate gauge complement	$2g$
$\mathcal{V}_{\text{log}}$ (V_{log})	Logical sector: $W_{\mathcal{G}}^{\perp}/W_{\mathcal{G}}$	$2k$
H_{stab}	Stabilizer-fixing subgroup of $\text{Sp}(2m, \mathbb{F}_2)$ (Def. 5)	$2^{r(r+1)/2}$
H_{gauge}	Gauge-normalizing subgroup (Def. 5)	Eq. (21)
$\text{Sp}(2g, \mathbb{F}_2)$	Symplectic group on gauge space	$2^{g^2} \prod_{i=1}^{g}(4^i - 1)$
$\text{Sol}(U_L)$	Set of physical realizations of logical U_L	$\lvert H_{\text{gauge}} \rvert$
$T_{\mathbf{h}}$	Symplectic transvection along $\mathbf{h}$	—
$\sigma(N)$	Lift of $N \in \text{Sp}(2g, \mathbb{F}_2)$ to H_{gauge}	Eq. (24)

II. Background

Notation. Table I summarizes the key symbols used throughout the paper. Vector spaces are over the binary field $\mathbb{F}_2$ with symplectic inner product $\langle \mathbf{u}, \mathbf{v} \rangle_s = \mathbf{u}\Omega\mathbf{v}^T$ where $\Omega = \left[\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right]$ (over $\mathbb{F}_2$, where $-1 = 1$, this is the standard alternating bilinear form). Isotropic subspaces (self-orthogonal under $\langle \cdot, \cdot \rangle_s$) are denoted $\mathcal{W}_S$; non-degenerate symplectic subspaces carry an induced non-singular restriction of this form.

A. The Binary Symplectic Representation

We summarize the binary symplectic representation of the Pauli and Clifford groups; see [3], [18] for full details.

For m physical qubits ($N = 2^m$-dimensional Hilbert space), the single-qubit Pauli matrices are $I, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. We define the displacement operator

$$D(\mathbf{a}, \mathbf{b}) = i^{\mathbf{a} \cdot \mathbf{b}} X^{a_1} Z^{b_1} \otimes \cdots \otimes X^{a_m} Z^{b_m} \quad (1)$$

for $\mathbf{a}, \mathbf{b} \in \mathbb{F}_2^m$. The Heisenberg-Weyl group $\mathcal{HW}_N$ consists of all $i^\kappa D(\mathbf{a}, \mathbf{b})$ with $\kappa \in \mathbb{Z}_4$. The homomorphism $\gamma : \mathcal{HW}_N \to \mathbb{F}_2^{2m}$,

$$\gamma(i^\kappa D(\mathbf{a}, \mathbf{b})) = [\mathbf{a} \,|\, \mathbf{b}], \quad (2)$$

has kernel $\langle iI_N \rangle$, allowing Paulis (up to phase) to be identified with binary vectors in $\mathbb{F}_2^{2m}$.

Definition 1 (Symplectic Inner Product)**.** *The* symplectic inner product *on* $\mathbb{F}_2^{2m}$ *is*

$$\langle \mathbf{u}, \mathbf{v} \rangle_s = \mathbf{u}\Omega\mathbf{v}^T \pmod 2, \quad \Omega = \begin{bmatrix} 0 & I_m \\ I_m & 0 \end{bmatrix}, \quad (3)$$

where $\mathbf{u} = [\mathbf{a}|\mathbf{b}]$, $\mathbf{v} = [\mathbf{a}'|\mathbf{b}']$. *Explicitly,* $\langle \mathbf{u}, \mathbf{v}\rangle_s = \mathbf{a}\mathbf{b}'^T + \mathbf{b}\mathbf{a}'^T \pmod 2$*. Two Paulis commute if and only if their symplectic inner product vanishes.*

The *symplectic group* $\mathrm{Sp}(2m, \mathbb{F}_2)$ is the group of all $2m \times 2m$ binary matrices M satisfying

$$M\Omega M^T = \Omega \pmod 2. \tag{4}$$

Every Clifford operator $F \in \mathcal{C}^{(2)}$ acts on $\mathcal{HW}_N$ by conjugation, inducing a well-defined linear map on $\mathbb{F}_2^{2m}$ that preserves the symplectic inner product; hence F is represented by $M_F \in \mathrm{Sp}(2m, \mathbb{F}_2)$ [19]. The elementary generators of $\mathrm{Sp}(2m, \mathbb{F}_2)$ correspond to CNOT, Hadamard, and Phase gates:

$$\begin{aligned} \mathrm{CNOT}_{i\to j}: &\quad \mathbf{e}_{X,i} \mapsto \mathbf{e}_{X,i} + \mathbf{e}_{X,j},\ \mathbf{e}_{Z,j} \mapsto \mathbf{e}_{Z,i} + \mathbf{e}_{Z,j}, & (5)\\ H_i: &\quad \mathbf{e}_{X,i} \leftrightarrow \mathbf{e}_{Z,i}, & (6)\\ P_i: &\quad \mathbf{e}_{X,i} \mapsto \mathbf{e}_{X,i} + \mathbf{e}_{Z,i},\ \mathbf{e}_{Z,i} \mapsto \mathbf{e}_{Z,i}. & (7) \end{aligned}$$

The order of the symplectic group is [20]:

$$|\mathrm{Sp}(2m, \mathbb{F}_2)| = 2^{m^2} \prod_{i=1}^{m} (4^i - 1). \tag{8}$$

Circuit synthesis reduces to constructing M_F from these generators via the Bruhat decomposition of $\mathrm{Sp}(2m, \mathbb{F}_2)$ [17].

B. Symplectic Transvections

A key tool in both the original and extended LCS theories is the symplectic transvection.

Definition 2 (Symplectic Transvection)**.** *For* $\mathbf{h} \in \mathbb{F}_2^{2m} \setminus \{0\}$*, the* symplectic transvection $T_{\mathbf{h}}$ *is the linear map*

$$T_{\mathbf{h}}(\mathbf{v}) = \mathbf{v} + \langle \mathbf{v}, \mathbf{h}\rangle_s\, \mathbf{h}. \tag{9}$$

$T_{\mathbf{h}}$ is a symplectic involution ($T_{\mathbf{h}}^2 = \mathrm{id}$), fixes the hyperplane $\mathbf{h}^\perp$ pointwise, and maps $\mathbf{h}$ to $\mathbf{h}$ (since $\langle \mathbf{h}, \mathbf{h}\rangle_s = 0$ over $\mathbb{F}_2$). Every element of $\mathrm{Sp}(2m, \mathbb{F}_2)$ is a product of transvections [20]. The stabilizer sector H_{stab} (defined precisely in Section III) is generated by $\{T_{\mathbf{s}} : \mathbf{s} \in W_{\mathcal{S}}\}$ (Theorem 1).

C. Stabilizer Codes and the LCS Theorem

An $[[m, k]]$ stabilizer code $\mathcal{C}$ is defined by an isotropic subspace $W_{\mathcal{S}} \subseteq \mathbb{F}_2^{2m}$ of dimension $r = m - k$. The stabilizer group $\mathcal{S}$ is the Abelian group generated by the Paulis $\{i^\kappa D(\mathbf{a}, \mathbf{b}) : [\mathbf{a}|\mathbf{b}] \in W_{\mathcal{S}}\}$. Logical Pauli operators form a symplectic basis $\{[\bar{X}_j], [\bar{Z}_j]\}_{j=1}^k$ of the quotient $W_{\mathcal{S}}^\perp / W_{\mathcal{S}}$:

$$\langle [\bar{X}_j], [\bar{Z}_\ell]\rangle_s = \delta_{j\ell}, \quad \langle [\bar{X}_j], W_{\mathcal{S}}\rangle_s = \langle [\bar{Z}_j], W_{\mathcal{S}}\rangle_s = 0. \tag{10}$$

Theorem 1 (LCS Theorem [3, Thm. 1])**.** *Let* $\mathcal{C}$ *be an* $[[m, k]]$ *stabilizer code with* $r = m - k$*. For every logical Clifford operator* U_L *on* k *qubits, the set of physical symplectic matrices* $M \in \mathrm{Sp}(2m, \mathbb{F}_2)$ *realizing* U_L *forms a left coset of the group*

$$H_{\mathrm{stab}}^{(m,k)} = \big\{ M \in \mathrm{Sp}(2m, \mathbb{F}_2) : M\mathbf{s} = \mathbf{s}\ \forall\, \mathbf{s} \in W_{\mathcal{S}},\ M|_{W_{\mathcal{S}}^\perp / W_{\mathcal{S}}} = \mathrm{id} \big\}, \tag{11}$$

which has order $|H_{\mathrm{stab}}^{(m,k)}| = 2^{r(r+1)/2}$.

The proof in [3] constructs $H_{\mathrm{stab}}^{(m,k)}$ explicitly as the group generated by transvections $\{T_{\mathbf{s}_i}\}$ for a basis $\{\mathbf{s}_i\}$ of $W_{\mathcal{S}}$, together with all products thereof. The count $2^{r(r+1)/2}$ equals the number of $r \times r$ upper triangular binary matrices with ones on the diagonal—equivalently, the order of the unipotent radical of the Borel subgroup of $\mathrm{Sp}(2r, \mathbb{F}_2)$.

Remark 1 (Scope of Theorem 1)**.** *The LCS theorem applies only when* $W_{\mathcal{S}}$ *is isotropic (i.e.,* $\langle \mathbf{s}, \mathbf{s}'\rangle_s = 0$ *for all* $\mathbf{s}, \mathbf{s}' \in W_{\mathcal{S}}$*). For a subsystem code with gauge space* $W_{\mathcal{G}}$*, the subspace* $W_{\mathcal{G}}$ *is* not *isotropic when* $g \geq 1$*: it contains pairs of anticommuting vectors corresponding to conjugate gauge generators. Applying the LCS theorem with parameter* $r' = m - k$ *(conflating gauge and stabilizer generators) is therefore incorrect; the theorem's hypotheses are not satisfied.*

D. Subsystem Stabilizer Codes

Definition 3 (Subsystem Stabilizer Code)**.** *An* $[[m, k, g, d]]$ subsystem stabilizer code *is defined by a (possibly non-Abelian) subgroup* $\mathcal{G} \leq \mathcal{HW}_N / \{\pm I, \pm iI\}$*, the* gauge group*, satisfying:*

- (i) *The center (stabilizer group)* $\mathcal{S} = Z(\mathcal{G})$ *is Abelian, with binary image* $W_{\mathcal{S}} = \gamma(\mathcal{S})$ *of dimension* $m - k - g$*;*
- (ii) *The gauge space* $W_{\mathcal{G}} = \gamma(\mathcal{G}) \subseteq \mathbb{F}_2^{2m}$ *has dimension* $m - k + g$*, so* $W_{\mathcal{S}} \subsetneq W_{\mathcal{G}}$*;*
- (iii) g *gauge qubits span* $\dim(W_{\mathcal{G}}/W_{\mathcal{S}}) = 2g$*;*
- (iv) *The code space is the joint* $+1$ *eigenspace of all elements of* $\mathcal{S}$*, and the code distance* d *is the minimum weight of a logical Pauli not in* $\mathcal{G}$*.*

We set $r = m - k - g = \dim W_{\mathcal{S}}$ throughout. Key dimensions:

$$\begin{aligned} \dim W_{\mathcal{G}} &= m - k + g = r + 2g, \\ \dim W_{\mathcal{G}}^\perp &= m - (r + 2g) = k + g, \\ \dim(W_{\mathcal{G}}^\perp / W_{\mathcal{G}}) &= 2k. \end{aligned} \tag{12}$$

The logical sector $V_{\log} = W_{\mathcal{G}}^\perp / W_{\mathcal{G}}$ has dimension $2k$ and carries a non-degenerate symplectic form induced from $\langle \cdot, \cdot \rangle_s$.

Remark 2 (Syndrome structure)**.** *A major practical advantage of subsystem codes is that syndrome extraction uses gauge generators (low-weight), not stabilizer generators. For the Bacon-Shor code, gauge generators are weight-2 while stabilizers are weight-4, reducing ancilla overhead and improving fault-tolerance [14].*

A physical Clifford F implements the logical action U_L on $\mathcal{C}$ if and only if its symplectic matrix M_F satisfies:

- **(G)** (*Gauge normalization*) $M_F W_{\mathcal{G}} = W_{\mathcal{G}}$;
- **(S)** (*Stabilizer centralization*) $M_F \mathbf{s} = \mathbf{s}$ for all $\mathbf{s} \in W_{\mathcal{S}}$;
- **(L)** (*Correct logical action*) M_F acts as U_L on the logical sector $V_{\log} = W_{\mathcal{G}}^\perp / W_{\mathcal{G}}$.

Condition **(G)** is strictly weaker than requiring M_F to fix $W_{\mathcal{G}}$ element-wise: normalization allows M_F to permute gauge-group elements non-trivially, as long as the subspace $W_{\mathcal{G}}$ is preserved as a set. Condition **(S)** is then the constraint that among these permutations, the stabilizer generators are fixed pointwise. The extra freedom lives entirely in the quotient $W_{\mathcal{G}}/W_{\mathcal{S}}$.

III. Logical Clifford Synthesis for Subsystem Codes

A. Problem Formulation

Definition 4 (Subsystem LCS Problem)**.** *Given an $[[m,k,g,d]]$ subsystem code with gauge space $W_{\mathcal{G}}$ and stabilizer space $W_{\mathcal{S}} \subseteq W_{\mathcal{G}}$, and a logical Clifford $U_L \in \mathcal{C}^{(2)}$ on k qubits, find all $M \in \mathrm{Sp}(2m,\mathbb{F}_2)$ satisfying conditions **(G)**, **(S)**, **(L)**.*

Definition 5 (Kernel Groups)**.** *Let $V_{\log} = W_{\mathcal{G}}^{\perp}/W_{\mathcal{G}}$ denote the $2k$-dimensional logical sector, and let $r = m-k-g$.*

$$H_{\mathrm{stab}} = \big\{M \in \mathrm{Sp}(2m,\mathbb{F}_2) : M\mathbf{s} = \mathbf{s}\ \forall \mathbf{s} \in W_{\mathcal{S}},\ M\mathbf{v} = \mathbf{v}\ \forall \mathbf{v} \in W_{\mathcal{G}},\ M|_{V_{\log}} = \mathrm{id}\big\}, \quad (13)$$

$$H_{\mathrm{gauge}} = \big\{M \in \mathrm{Sp}(2m,\mathbb{F}_2) : MW_{\mathcal{G}} = W_{\mathcal{G}},\ M\mathbf{s} = \mathbf{s}\ \forall \mathbf{s} \in W_{\mathcal{S}},\ M|_{V_{\log}} = \mathrm{id}\big\}. \quad (14)$$

Remark 3 (Disambiguation from Theorem 1)**.** *The group H_{stab} of Definition 5 differs from $H_{\mathrm{stab}}^{(m,k)}$ of Theorem 1 in two respects. First, H_{stab} uses $r = m-k-g$ (only the true stabilizer rank), whereas $H_{\mathrm{stab}}^{(m,k)}$ uses $r = m-k$ (the full syndrome rank including gauge). Second, H_{stab} imposes the additional condition $M\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in W_{\mathcal{G}}$, fixing the gauge space pointwise; $H_{\mathrm{stab}}^{(m,k)}$ only fixes $W_{\mathcal{S}}$. These distinctions are essential: $H_{\mathrm{stab}} \le H_{\mathrm{gauge}}$, and $|H_{\mathrm{stab}}| = 2^{r(r+1)/2}$ with $r = m-k-g$ by Theorem 1 applied to the gauge-fixed stabilizer code with parameters $[[m, k+g]]$.*

The inclusions $H_{\mathrm{stab}} \le H_{\mathrm{gauge}} \le \mathrm{Sp}(2m,\mathbb{F}_2)$ are immediate from the definitions.

Lemma 2 (Existence and Coset Structure)**.** *For every logical Clifford U_L on the subsystem code, there exists at least one $M_0 \in \mathrm{Sp}(2m,\mathbb{F}_2)$ satisfying conditions **(G)**, **(S)**, **(L)**, and the complete solution set is*

$$\mathrm{Sol}(U_L) = M_0 \cdot H_{\mathrm{gauge}}. \quad (15)$$

Proof. **Existence.** The logical sector $V_{\log} = W_{\mathcal{G}}^{\perp}/W_{\mathcal{G}}$ is a non-degenerate symplectic space of dimension $2k$. The map $U_L \mapsto M_{U_L} \in \mathrm{Sp}(2k,\mathbb{F}_2)$ gives a symplectic automorphism of $V_{\log}$. By [20, Thm. 8.1], every isometry of a non-degenerate symplectic subspace of $\mathbb{F}_2^{2m}$ extends to a symplectic isometry of the full space. Explicitly: choose any complement $V'_{\log}$ to $W_{\mathcal{G}}^{\perp}$ in $\mathbb{F}_2^{2m}$ (equivalently, a choice of representatives for the coset space), and extend M_{U_L} to act as the identity on $W_{\mathcal{G}}$ and on $V'_{\log}$. The resulting $M_0 \in \mathrm{Sp}(2m,\mathbb{F}_2)$ satisfies: **(G)** since M_0 fixes $W_{\mathcal{G}}$ pointwise; **(S)** since $W_{\mathcal{S}} \subseteq W_{\mathcal{G}}$; **(L)** by construction.

Coset structure. Suppose $M_0, M_1 \in \mathrm{Sol}(U_L)$. Set $N = M_0^{-1}M_1$. Since M_0 and M_1 both satisfy **(G)**, so does N (the stabilizer of the subspace $W_{\mathcal{G}}$ in $\mathrm{Sp}(2m,\mathbb{F}_2)$ is a group). Since both fix $W_{\mathcal{S}}$ pointwise, so does N. Since both act as U_L on $V_{\log}$, the composition acts trivially: $N|_{V_{\log}} = \mathrm{id}$. Hence $N \in H_{\mathrm{gauge}}$.

Conversely, for any $h \in H_{\mathrm{gauge}}$, the product $M_0 h$ satisfies: $M_0 h\, W_{\mathcal{G}} = M_0 W_{\mathcal{G}} = W_{\mathcal{G}}$ (**G**); for $\mathbf{s} \in W_{\mathcal{S}}$, $M_0 h\,\mathbf{s} = M_0\mathbf{s} = \mathbf{s}$ since h fixes $W_{\mathcal{S}}$ (**S**); and $M_0 h$ acts as $U_L \cdot \mathrm{id} = U_L$ on $V_{\log}$ (**L**). Hence $\mathrm{Sol}(U_L) = M_0 \cdot H_{\mathrm{gauge}}$. □

Remark 4. *The solution set is a* left *coset in our convention because M_0 acts first (circuit concatenation reads right-to-left but matrices left-multiply column vectors). The set $\mathrm{Sol}(U_L)$ has the same cardinality as H_{gauge} regardless of U_L.*

B. Symplectic Decomposition of the Gauge Space

Lemma 3 (Gauge Space Decomposition)**.** *For any $[[m,k,g,d]]$ subsystem stabilizer code with Abelian center $\mathcal{S} = Z(\mathcal{G})$, there exists a subspace $V_{\mathrm{gauge}} \subseteq W_{\mathcal{G}}$ such that*

$$W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\mathrm{gauge}} \quad (16)$$

(direct sum of $\mathbb{F}_2$-vector spaces), where $W_{\mathcal{S}}$ is isotropic, V_{gauge} is non-degenerate symplectic of dimension $2g$, and $\langle \mathbf{s}, \mathbf{v}\rangle_s = 0$ for all $\mathbf{s} \in W_{\mathcal{S}}$, $\mathbf{v} \in V_{\mathrm{gauge}}$. Moreover, V_{gauge} is orthogonal to $V_{\log}$ in the symplectic sense:

$$\langle \mathbf{v}, \mathbf{w}\rangle_s = 0 \quad \forall\, \mathbf{v} \in V_{\mathrm{gauge}},\ \mathbf{w} \in V_{\log}. \quad (17)$$

Proof. The restriction of the symplectic form ω of $\mathbb{F}_2^{2m}$ to $W_{\mathcal{G}}$ has radical $W_{\mathcal{G}} \cap W_{\mathcal{G}}^{\perp} = W_{\mathcal{S}}$. To see this: $\mathbf{v} \in W_{\mathcal{G}}$ satisfies $\langle \mathbf{v}, \mathbf{w}\rangle_s = 0$ for all $\mathbf{w} \in W_{\mathcal{G}}$ if and only if $D(\mathbf{v})$ commutes with every element of $\mathcal{G}$, i.e., $D(\mathbf{v})$ lies in the center $Z(\mathcal{G}) = \mathcal{S}$, hence $\mathbf{v} \in W_{\mathcal{S}}$. Thus $\mathrm{rad}(\omega|_{W_{\mathcal{G}}}) = W_{\mathcal{S}}$.

The quotient $W_{\mathcal{G}}/W_{\mathcal{S}}$ therefore carries a non-degenerate symplectic form $\bar{\omega}$ induced by ω. Since every non-degenerate symplectic space over $\mathbb{F}_2$ is even-dimensional and admits a symplectic basis [20], there exist $\bar{e}_1, \bar{f}_1, \ldots, \bar{e}_g, \bar{f}_g \in W_{\mathcal{G}}/W_{\mathcal{S}}$ with $\bar{\omega}(\bar{e}_i, \bar{f}_j) = \delta_{ij}$ and $\bar{\omega}(\bar{e}_i, \bar{e}_j) = \bar{\omega}(\bar{f}_i, \bar{f}_j) = 0$. Lift these to vectors $e_1, f_1, \ldots, e_g, f_g \in W_{\mathcal{G}}$ and set $V_{\mathrm{gauge}} = \mathrm{span}\{e_1, f_1, \ldots, e_g, f_g\}$.

Trivial intersection. Any $\mathbf{s} \in W_{\mathcal{S}} \cap V_{\mathrm{gauge}}$ maps to 0 in $W_{\mathcal{G}}/W_{\mathcal{S}}$, so $\mathbf{s} = 0$. Hence $W_{\mathcal{S}} \cap V_{\mathrm{gauge}} = \{0\}$. Dimension counting gives $\dim(W_{\mathcal{S}} \oplus V_{\mathrm{gauge}}) = r + 2g = \dim W_{\mathcal{G}}$, so $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\mathrm{gauge}}$.

Orthogonality $W_{\mathcal{S}} \perp V_{\mathrm{gauge}}$. Each e_i, f_j was lifted from $W_{\mathcal{G}}/W_{\mathcal{S}}$, and the form $\bar{\omega}$ on $W_{\mathcal{G}}/W_{\mathcal{S}}$ is the restriction of ω. The isotropicity of $W_{\mathcal{S}}$ gives $\langle \mathbf{s}, \mathbf{v}\rangle_s = 0$ for all $\mathbf{s} \in W_{\mathcal{S}}$ and $\mathbf{v} \in V_{\mathrm{gauge}}$ since $\mathbf{s}$ maps to 0 in $W_{\mathcal{G}}/W_{\mathcal{S}}$ and hence to the radical of $\bar{\omega}$.

Orthogonality $V_{\mathrm{gauge}} \perp V_{\log}$. Every representative $\mathbf{w}$ of a class in $V_{\log}$ lies in $W_{\mathcal{G}}^{\perp}$ (by definition of $W_{\mathcal{G}}^{\perp}$). Since $V_{\mathrm{gauge}} \subseteq W_{\mathcal{G}}$, we have $\langle \mathbf{w}, \mathbf{v}\rangle_s = 0$ for all $\mathbf{v} \in V_{\mathrm{gauge}}$ and all $\mathbf{w} \in W_{\mathcal{G}}^{\perp}$. □

Lemma 3 implies a global orthogonal decomposition of $\mathbb{F}_2^{2m}$ in the symplectic sense:

$$\mathbb{F}_2^{2m} = V_{\mathrm{gauge}} \perp V_{\mathrm{gauge}}^{\perp}, \quad (18)$$

where $V_{\mathrm{gauge}}^{\perp}$ contains $W_{\mathcal{S}}$, the logical sector $V_{\log}$, and a remaining complement V_{rem}:

$$V_{\mathrm{gauge}}^{\perp} = W_{\mathcal{S}} \oplus V_{\log} \oplus V_{\mathrm{rem}},$$
$$\dim V_{\mathrm{rem}} = 2(m-k-r-g) = 0 \ \text{ iff } \ m = k+r+g. \quad (19)$$

In the standard case $m = k + r + g$ (i.e., $r + g + k = m$, which holds by definition since $r = m - k - g$), we have $V_{\text{rem}} = \{0\}$ and the decomposition $V_{\text{gauge}}^{\perp} = W_{\mathcal{S}} \oplus V_{\log}$ is exact. The dimension count is $\dim V_{\text{gauge}} + \dim V_{\text{gauge}}^{\perp} = 2g + (2m - 2g) = 2m$, consistent.

C. Main Theorem

Theorem 4 (Product Structure of Solutions)**.** *Let $\mathcal{C}$ be an $[[m, k, g, d]]$ subsystem stabilizer code with Abelian center $\mathcal{S} = Z(\mathcal{G})$ and $r = m - k - g$. The group H_{gauge} of Definition 5 splits as a direct product:*

$$H_{\text{gauge}} \cong H_{\text{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2). \tag{20}$$

Consequently, for every logical Clifford operator U_L on the code:

$$|\mathrm{Sol}(U_L)| = 2^{r(r+1)/2} \cdot |\mathrm{Sp}(2g, \mathbb{F}_2)|, \tag{21}$$

where $|\mathrm{Sp}(2g, \mathbb{F}_2)| = 2^{g^2} \prod_{i=1}^{g}(4^i - 1)$. When $g = 0$, equation (21) reduces to $2^{r(r+1)/2}$ with $r = m - k$, recovering Theorem 1.

Proof. Fix the orthogonal decomposition $\mathbb{F}_2^{2m} = V_{\text{gauge}} \perp V_{\text{gauge}}^{\perp}$ of Lemma 3, and define the restriction homomorphism

$$\varphi : H_{\text{gauge}} \longrightarrow \mathrm{Sp}(2g, \mathbb{F}_2), \qquad M \longmapsto M|_{V_{\text{gauge}}}. \tag{22}$$

(1) φ is a well-defined homomorphism. Let $M \in H_{\text{gauge}}$. We must show $M(V_{\text{gauge}}) = V_{\text{gauge}}$ and that $M|_{V\,\text{gauge}}$ preserves the symplectic form.

Since $MW_{\mathcal{G}} = W_{\mathcal{G}}$ (condition **G**) and $M|_{W_{\mathcal{S}}} = \text{id}$ (condition **S**), for any $\mathbf{v} \in V_{\text{gauge}}$ write $M\mathbf{v} = \mathbf{s} + \mathbf{v}'$ with $\mathbf{s} \in W_{\mathcal{S}}$ and $\mathbf{v}' \in V_{\text{gauge}}$ (using $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\text{gauge}}$). For any $\mathbf{t} \in W_{\mathcal{S}}$:

$$\langle \mathbf{s}, \mathbf{t} \rangle_s = \langle M\mathbf{v} - \mathbf{v}', \mathbf{t} \rangle_s = \langle M\mathbf{v}, M\mathbf{t} \rangle_s - \langle \mathbf{v}', \mathbf{t} \rangle_s = 0 - 0 = 0, \tag{23}$$

where we used that M is symplectic (preserving the inner product), $M\mathbf{t} = \mathbf{t}$, and $V_{\text{gauge}} \perp W_{\mathcal{S}}$. Since $W_{\mathcal{S}}$ is non-degenerately paired with a complementary space, $\langle \mathbf{s}, \mathbf{t} \rangle_s = 0$ for all $\mathbf{t} \in W_{\mathcal{S}}$ forces $\mathbf{s} = 0$ (as $W_{\mathcal{S}}$ is isotropic, but $\mathbf{s}$ is also in $W_{\mathcal{S}}$, and $\langle \mathbf{s}, \cdot \rangle_s$ restricted to $W_{\mathcal{G}}^{\perp}$ determines $\mathbf{s}$ uniquely).

More precisely: the form $\langle \cdot, \cdot \rangle_s$ pairs $W_{\mathcal{S}}$ injectively with $W_{\mathcal{G}}^{\perp}$ (since $W_{\mathcal{S}} \cap W_{\mathcal{G}}^{\perp} = \{0\}$ by $r = m - k - g$ and the code being non-degenerate). For $\mathbf{s} \in W_{\mathcal{S}}$, $\langle \mathbf{s}, \mathbf{u} \rangle_s = 0$ for all $\mathbf{u} \in W_{\mathcal{S}}$ (isotropic) and for all $\mathbf{u} \in V_{\text{gauge}}$ (Lemma 3), hence for all $\mathbf{u} \in W_{\mathcal{G}}$. Since M is symplectic and $MW_{\mathcal{G}} = W_{\mathcal{G}}$, we also have $\langle \mathbf{s}, \mathbf{u} \rangle_s = 0$ for all $\mathbf{u}$ in the complement of $W_{\mathcal{G}}^{\perp}$. Therefore $\mathbf{s} = 0$, giving $M\mathbf{v} \in V_{\text{gauge}}$. By dimension, $M|_{V\,\text{gauge}}$ is bijective.

Since V_{gauge} is a non-degenerate symplectic subspace orthogonal to $V_{\text{gauge}}^{\perp}$ and M is symplectic on $\mathbb{F}_2^{2m}$, the restriction $M|_{V\,\text{gauge}}$ preserves $\langle \cdot, \cdot \rangle_s|_{V_{\text{gauge}}}$, giving $M|_{V\,\text{gauge}} \in \mathrm{Sp}(2g, \mathbb{F}_2)$.

The homomorphism property holds since $\varphi(M_1M_2) = (M_1M_2)|_{V\,\text{gauge}} = M_1|_{V\,\text{gauge}} \cdot M_2|_{V\,\text{gauge}} = \varphi(M_1)\,\varphi(M_2)$.

(2) $\ker\varphi = H_{\text{stab}}$. Suppose $M \in \ker\varphi$, i.e., $M \in H_{\text{gauge}}$ and $M|_{V\,\text{gauge}} = \text{id}$. Then M fixes $W_{\mathcal{S}}$ pointwise (condition **S**) and V_{gauge} pointwise (kernel condition), hence fixes all of $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\text{gauge}}$ pointwise. Together with condition **(L)** inherited from $M \in H_{\text{gauge}}$, which gives $M|_{V_{\log}} = \text{id}$, we conclude $M \in H_{\text{stab}}$ per Definition 5.

Conversely, every $M \in H_{\text{stab}}$ fixes $W_{\mathcal{G}}$ pointwise (in particular V_{gauge}), so $M \in \ker\varphi$. Hence $\ker\varphi = H_{\text{stab}}$.

(3) φ is surjective. For any $N \in \mathrm{Sp}(2g, \mathbb{F}_2)$, define the lift

$$\sigma(N) : \mathbb{F}_2^{2m} \longrightarrow \mathbb{F}_2^{2m},$$
$$\sigma(N)\big|_{V_{\text{gauge}}} = N, \quad \sigma(N)\big|_{V_{\text{gauge}}^{\perp}} = \text{id}. \tag{24}$$

Since $V_{\text{gauge}} \perp V_{\text{gauge}}^{\perp}$ symplectically and N is symplectic on V_{gauge}, the block-diagonal map $\sigma(N)$ satisfies $\sigma(N)\Omega\sigma(N)^T = \Omega$ globally, hence $\sigma(N) \in \mathrm{Sp}(2m, \mathbb{F}_2)$. The three conditions are satisfied:

- **(G):** $\sigma(N)W_{\mathcal{G}} = W_{\mathcal{S}} \oplus N(V_{\text{gauge}}) = W_{\mathcal{S}} \oplus V_{\text{gauge}} = W_{\mathcal{G}}$;
- **(S):** $\sigma(N)|_{W_{\mathcal{S}}} = \text{id}$, since $W_{\mathcal{S}} \subseteq V_{\text{gauge}}^{\perp}$;
- **(L):** $\sigma(N)|_{V_{\log}} = \text{id}$, since $V_{\log} \subseteq V_{\text{gauge}}^{\perp}$.

Hence $\sigma(N) \in H_{\text{gauge}}$ and $\varphi(\sigma(N)) = N$, so φ is surjective.

(4) The sequence splits. The section $\sigma : \mathrm{Sp}(2g, \mathbb{F}_2) \to H_{\text{gauge}}$ satisfies $\sigma(N_1)\sigma(N_2) = \sigma(N_1N_2)$ since both sides act as N_1N_2 on V_{gauge} and as id on $V_{\text{gauge}}^{\perp}$. Hence σ is a group homomorphism with $\varphi \circ \sigma = \text{id}$, so the short exact sequence

$$1 \longrightarrow H_{\text{stab}} \longrightarrow H_{\text{gauge}} \xrightarrow{\varphi} \mathrm{Sp}(2g, \mathbb{F}_2) \longrightarrow 1 \tag{25}$$

splits, giving $H_{\text{gauge}} \cong H_{\text{stab}} \rtimes \mathrm{Sp}(2g, \mathbb{F}_2)$.

(5) The semidirect product is direct. It remains to show H_{stab} is central in H_{gauge}, i.e., $\sigma(N)\,h\,\sigma(N)^{-1} = h$ for all $h \in H_{\text{stab}}$ and $N \in \mathrm{Sp}(2g, \mathbb{F}_2)$.

On V_{gauge}: since $h \in H_{\text{stab}}$ fixes V_{gauge} pointwise (Definition 5), we have $\sigma(N)\,h\,\sigma(N)^{-1}|_{V\,\text{gauge}} = N \circ \text{id} \circ N^{-1} = \text{id} = h|_{V\,\text{gauge}}$.

On $V_{\text{gauge}}^{\perp}$: since $\sigma(N)|_{V_{\text{gauge}}^{\perp}} = \text{id}$, we have $\sigma(N)\,h\,\sigma(N)^{-1}|_{V_{\text{gauge}}^{\perp}} = h|_{V_{\text{gauge}}^{\perp}}$.

Therefore $\sigma(N)\,h\,\sigma(N)^{-1} = h$, so H_{stab} is central in H_{gauge} and the semidirect product is direct:

$$H_{\text{gauge}} \cong H_{\text{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2). \tag{26}$$

The count (21) follows from $|H_{\text{stab}}| = 2^{r(r+1)/2}$ (Remark 3 and Theorem 1) and (8) with $m \to g$. ☐

Corollary 5 (Uniformity)**.** *The count $|\mathrm{Sol}(U_L)|$ is the same for every logical Clifford U_L on a given subsystem code.*

Proof. By Lemma 2, the solution sets for distinct U_L's are distinct left cosets of H_{gauge} in $\mathrm{Sp}(2m, \mathbb{F}_2)$, all of cardinality $|H_{\text{gauge}}|$. ☐

Corollary 6 (Separable Optimization)**.** *The solution space $\mathrm{Sol}(U_L) = M_0 \cdot (H_{\text{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2))$ factors as a Cartesian product. Optimization over stabilizer degrees of freedom (first factor) is independent of optimization over gauge degrees of freedom (second factor).*

Proof. Immediate from the direct product structure (20). For any objective function $f : \mathrm{Sp}(2m, \mathbb{F}_2) \to \mathbb{R}$ that decomposes as $f(M_0 \cdot h \cdot \sigma(N)) = f_{\text{stab}}(h) + f_{\text{gauge}}(N)$ (e.g., two-qubit gate count when stabilizer and gauge subcircuits act on disjoint qubits), the joint optimum is attained by separate optimization:

$$\min_{h,N} f = \min_{h} f_{\text{stab}}(h) + \min_{N} f_{\text{gauge}}(N). \tag{27}$$

$\square$

Remark 5 (Non-Abelian gauge groups and restricted gauge factors)**.** *If $\mathcal{G}$ is non-Abelian but $Z(\mathcal{G})$ is still a well-defined stabilizer group (as in all subsystem stabilizer codes by definition), the decomposition $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\text{gauge}}$ of Lemma 3 remains valid, since it relies only on the radical structure of the symplectic form restricted to $W_{\mathcal{G}}$. The splitting section σ of equation (24) is likewise constructed purely from the orthogonal decomposition $\mathbb{F}_2^{2m} = V_{\text{gauge}} \perp V_{\text{gauge}}^{\perp}$, with no reference to the algebraic structure of $\mathcal{G}$. The short exact sequence (25) therefore* always *splits for any subsystem stabilizer code with well-defined $W_{\mathcal{S}} = \gamma(Z(\mathcal{G}))$.*

However, the image *of $\varphi : H_{\text{gauge}} \to \mathrm{Sp}(2g, \mathbb{F}_2)$ may be a proper subgroup when physical constraints restrict the allowed gauge permutations. The principal example is time-periodic (Floquet) gauge groups [7]: in the honeycomb Floquet code, the gauge group cycles through three distinct configurations under the measurement schedule, and only gauge transformations compatible with this periodicity are physically implementable. In such cases the gauge factor $|\mathrm{Sp}(2g, \mathbb{F}_2)|$ in equation (21) should be replaced by $|\mathrm{Im}(\varphi)| \le |\mathrm{Sp}(2g, \mathbb{F}_2)|$, and identifying $\mathrm{Im}(\varphi)$ precisely for specific Floquet codes is an important open problem. For all static subsystem stabilizer codes considered in this paper, no such constraint applies and φ is surjective (Theorem 4, step 3).*

Remark 6 (Recovery of the stabilizer LCS theorem)**.** *For $g = 0$, the subsystem code reduces to a stabilizer code with $W_{\mathcal{G}} = W_{\mathcal{S}}$, $V_{\text{gauge}} = \{0\}$, and $\mathrm{Sp}(0, \mathbb{F}_2) = \{1\}$. Equation (21) gives $|\mathrm{Sol}(U_L)| = 2^{r(r+1)/2}$ with $r = m - k$, recovering Theorem 1.*

IV. Algorithm

The direct product structure $H_{\text{gauge}} \cong H_{\text{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2)$ suggests a natural two-phase enumeration: first enumerate all stabilizer-sector solutions using the original LCS algorithm, then independently enumerate all gauge-sector symplectic matrices and lift them to the full space. Theorem 4 guarantees this yields all solutions. We formalize this as Algorithm 1, analyse its correctness and complexity, and describe circuit extraction via the Bruhat decomposition.

Algorithm 1 Subsystem Logical Clifford Synthesis (SubLCS)

Require: $[[m, k, g, d]]$ subsystem code $(W_{\mathcal{S}}, W_{\mathcal{G}})$ with $W_{\mathcal{S}} \subseteq W_{\mathcal{G}} \subseteq \mathbb{F}_2^{2m}$; target logical Clifford U_L

Ensure: All symplectic solutions $\mathrm{Sol}(U_L) \subseteq \mathrm{Sp}(2m, \mathbb{F}_2)$

```
// — Phase 1: Stabilizer solutions —
Compute a symplectic basis B = {s_1, ..., s_r, t_1, ..., t_r, x_1, z_1, ..., x_k, z_k, ...} for F_2^{2m} adapted to W_S, using the symplectic Gram-Schmidt procedure [21]
Construct the target matrix F_{U_L} ∈ Sp(2m, F_2) encoding the action of U_L on the logical sector, extended trivially to W_S and its complement
Call the LCS Algorithm of [3]: enumerate {M_1, ..., M_{2^{r(r+1)/2}}} ⊆ Sp(2m, F_2) (one base solution plus |H_stab| coset elements)
// — Phase 2: Gauge orbit —
Compute the gauge complement V_gauge: within W_G, apply symplectic Gram-Schmidt to W_G starting from W_S, extracting the orthogonal complement basis {e_1, f_1, ..., e_g, f_g} with ⟨e_i, f_j⟩_s = δ_ij
Identify the embedding ι : F_2^{2g} ↪ F_2^{2m} mapping the standard basis to {e_1, f_1, ..., e_g, f_g}
for each N ∈ Sp(2g, F_2) do
    Construct the lift: σ(N)|_{V gauge} = ι N ι^{-1}, σ(N)|_{V_gauge^⊥} = id (block-diagonal in the basis B)
end for
// — Combine —
return {M_i · σ(N) : 1 ≤ i ≤ 2^{r(r+1)/2}, N ∈ Sp(2g, F_2)}
```

A. Correctness

Proposition 7. *Algorithm 1 returns exactly $\mathrm{Sol}(U_L)$.*

Proof. By Lemma 2, $\mathrm{Sol}(U_L) = M_0 \cdot H_{\text{gauge}}$ for any base solution M_0. By Theorem 4, every element of H_{gauge} has the form $h \cdot \sigma(N)$ with $h \in H_{\text{stab}}$ and $N \in \mathrm{Sp}(2g, \mathbb{F}_2)$, and distinct pairs (h, N) yield distinct elements (direct product). Phase 1 computes $\{M_1, \ldots, M_{2^{r(r+1)/2}}\} = M_0 \cdot H_{\text{stab}}$ by the correctness of [3]. Phase 2 computes $\{\sigma(N) : N \in \mathrm{Sp}(2g, \mathbb{F}_2)\}$ as the image of the section σ. The output is $\{M_i \cdot \sigma(N)\} = M_0 \cdot H_{\text{stab}} \cdot \{\sigma(N)\} = M_0 \cdot H_{\text{gauge}} = \mathrm{Sol}(U_L)$. $\square$

B. Complexity Analysis

The two phases of Algorithm 1 are asymptotically independent: Phase 1 is dominated by the original LCS cost over the stabilizer sector, while Phase 2 adds overhead only in the gauge dimension g, which is typically much smaller than m.

Proposition 8 (Complexity)**.** *Algorithm 1 has the following running times:*

- Phase 1 (basis + stabilizer enumeration): $O(m^3)$ *operations over $\mathbb{F}_2$ to compute the symplectic basis; $O(m^2)$ per element of H_{stab}; total $O(m^3 + 2^{r(r+1)/2} \cdot m^2)$.*
- Phase 2 (gauge complement + lift): $O(g^3)$ *to compute V_{gauge} by symplectic Gram-Schmidt within $W_{\mathcal{G}}$; $O(m^2)$ per lift $\sigma(N)$ (constructing a $2m \times 2m$ block-diagonal matrix); total $O(g^3 + |\mathrm{Sp}(2g, \mathbb{F}_2)| \cdot m^2)$.*
- Combine: $O(m^2)$ *per output matrix (one $2m \times 2m$ matrix multiplication).*
- Total: $O(m^3)$ *preprocessing plus $O(m^2)$ per solution output.*

Note that the total number of solutions grows super-exponentially in g via $|\mathrm{Sp}(2g, \mathbb{F}_2)|$; for large g, full enumeration of the gauge orbit dominates. The algorithm supports restricting Phase 2 to a generator set or hardware-relevant subset of $\mathrm{Sp}(2g, \mathbb{F}_2)$.

C. Circuit Extraction

Each symplectic matrix $M \in \mathrm{Sol}(U_L)$ is converted to an explicit quantum circuit using the Bruhat decomposition of $\mathrm{Sp}(2m, \mathbb{F}_2)$ [17]:

$$M = L\,D\,U, \tag{28}$$

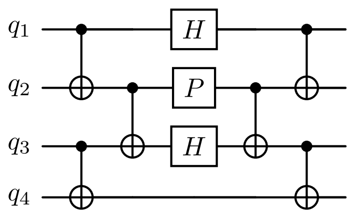

Fig. 1. Bruhat decomposition $M = LDU$ of a symplectic matrix $M \in \mathrm{Sp}(2m, \mathbb{F}_2)$ into a lower-unitriangular CNOT layer L (left block), a diagonal layer D of single-qubit Hadamard and Phase gates (centre), and an upper-unitriangular CNOT layer U (right block). Each solution $M \in \mathrm{Sol}(U_L)$ output by Algorithm 1 is converted to an explicit circuit via this decomposition, with $O(m^2)$ CNOT gates in the worst case [17]. The product structure of Corollary 6 means the stabilizer and gauge subcircuits occupy disjoint blocks of L, D, U and can be optimized independently.

where L is lower-unitriangular (CNOT layer), D is block-diagonal (Hadamards + Phases), and U is upper-unitriangular (CNOT layer). This decomposition yields $O(m^2)$ CNOT gates in the worst case and is optimal up to constants [17]. The product structure of Corollary 6 means that the stabilizer and gauge subcircuits can be extracted and optimized independently before combining (Fig. 1).

V. Example: The $[[4,1,1,2]]$ Bacon-Shor Code

The 2×2 Bacon-Shor code [5] has $m = 4$, $k = 1$, $g = 1$, $r = m - k - g = 2$, $d = 2$. We label the four physical qubits in row-major order: $1 = (1,1)$, $2 = (1,2)$, $3 = (2,1)$, $4 = (2,2)$.

A. Code Structure

Gauge generators:

$$g_{X,1} = X_1 X_2 = [1100 \,|\, 0000], \tag{29}$$
$$g_{X,2} = X_3 X_4 = [0011 \,|\, 0000], \tag{30}$$
$$g_{Z,1} = Z_1 Z_3 = [0000 \,|\, 1010], \tag{31}$$
$$g_{Z,2} = Z_2 Z_4 = [0000 \,|\, 0101]. \tag{32}$$

One verifies that $\langle g_{X,1}, g_{Z,1} \rangle_s = 1$ (they share qubit 1 in an XZ-overlap), so $\mathcal{G}$ is non-Abelian.
Stabilizers (center $Z(\mathcal{G})$):

$$S_X = g_{X,1} \cdot g_{X,2} = X_1 X_2 X_3 X_4 = [1111 \,|\, 0000], \tag{33}$$
$$S_Z = g_{Z,1} \cdot g_{Z,2} = Z_1 Z_2 Z_3 Z_4 = [0000 \,|\, 1111]. \tag{34}$$

One verifies $\langle S_X, S_Z \rangle_s = 0$ and that S_X, S_Z commute with all gauge generators. Thus $W_{\mathcal{S}} = \mathrm{span}\{[1111|0000], [0000|1111]\}$, $\dim W_{\mathcal{S}} = r = 2$.
Logical operators:

$$\bar{X}_L = X_1 X_3 = [1010 \,|\, 0000], \tag{35}$$
$$\bar{Z}_L = Z_1 Z_2 = [0000 \,|\, 1100]. \tag{36}$$

We verify: $\langle \bar{X}_L, \bar{Z}_L \rangle_s = 1$; $\langle \bar{X}_L, S_X \rangle_s = \langle \bar{X}_L, S_Z \rangle_s = 0$; $\langle \bar{Z}_L, S_X \rangle_s = \langle \bar{Z}_L, S_Z \rangle_s = 0$.
Gauge space:

$$W_{\mathcal{G}} = \mathrm{span}\big\{[1100|0000],\ [0011|0000],\ [0000|1010],\ [0000|0101]\big\}, \tag{37}$$

with $\dim W_{\mathcal{G}} = 4 = r + 2g = 2 + 2$, consistent with (12).

B. Symplectic Decomposition

We apply Lemma 3 to extract V_{gauge}.

Proposition 9. *The subspace $V_{\text{gauge}} = \mathrm{span}\{v_X, v_Z\}$ with*

$$v_X = [1100 \,|\, 0000], \quad v_Z = [0000 \,|\, 1010], \tag{38}$$

satisfies all conditions of Lemma 3 for the $[[4,1,1,2]]$ code.

Proof. We verify each condition of Lemma 3 in turn.

(1) Non-degeneracy.

$$\begin{aligned}\langle v_X, v_Z \rangle_s &= [1100] \cdot [1010]^T + [0000] \cdot [0000]^T \\ &= (1{\cdot}1 + 1{\cdot}0 + 0{\cdot}1 + 0{\cdot}0) \bmod 2 = 1,\end{aligned} \tag{39}$$

so V_{gauge} is a non-degenerate symplectic subspace with $g = 1$.

(2) Trivial intersection with $W_{\mathcal{S}}$. The stabilizers $[1111|0000]$ and $[0000|1111]$ have full support on all four qubits, while v_X and v_Z each have support on only two. Hence $W_{\mathcal{S}} \cap V_{\text{gauge}} = \{0\}$.

(3) Spanning. $\dim(W_{\mathcal{S}} \oplus V_{\text{gauge}}) = 2 + 2 = 4 = \dim W_{\mathcal{G}}$, and $W_{\mathcal{S}}, V_{\text{gauge}} \subseteq W_{\mathcal{G}}$, so $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\text{gauge}}$.

(4) Orthogonality $W_{\mathcal{S}} \perp V_{\text{gauge}}$.

$$\begin{aligned}\langle v_X, S_X \rangle_s &= [1100] \cdot [0000]^T + [0000] \cdot [1111]^T \\ &= 0,\end{aligned} \tag{40}$$
$$\begin{aligned}\langle v_X, S_Z \rangle_s &= [1100] \cdot [1111]^T + [0000] \cdot [0000]^T \\ &= (1{+}1{+}0{+}0) \bmod 2 = 0,\end{aligned} \tag{41}$$

and analogously $\langle v_Z, S_X \rangle_s = \langle v_Z, S_Z \rangle_s = 0$.

(5) Orthogonality $V_{\text{gauge}} \perp V_{\log}$.

$$\begin{aligned}\langle v_X, \bar{X}_L \rangle_s &= [1100] \cdot [0000]^T + [0000] \cdot [1010]^T \\ &= 0,\end{aligned} \tag{42}$$
$$\begin{aligned}\langle v_X, \bar{Z}_L \rangle_s &= [1100] \cdot [1100]^T + [0000] \cdot [0000]^T \\ &= (1{+}1{+}0{+}0) \bmod 2 = 0,\end{aligned} \tag{43}$$

and analogously $\langle v_Z, \bar{X}_L \rangle_s = \langle v_Z, \bar{Z}_L \rangle_s = 0$. □

The full symplectic basis of $\mathbb{F}_2^8$ adapted to this decomposition is:

$$\text{Gauge: } v_X = [1100|0000],\ v_Z = [0000|1010]; \tag{44}$$
$$\text{Stabilizer: } s_X = [1111|0000],\ s_Z = [0000|1111]; \tag{45}$$
$$\text{Logical: } \bar{X}_L = [1010|0000],\ \bar{Z}_L = [0000|1100]. \tag{46}$$

Since $r + 2g + 2k = 2 + 2 + 2 = 6 < 2m = 8$, there is a two-dimensional remainder subspace V_{rem}; a convenient choice is:

$$\text{Remainder: } r_X = [0001|0000],\ r_Z = [0000|0001]. \tag{47}$$

One verifies all required symplectic inner products among these eight basis vectors.

TABLE II
THE SIX ELEMENTS OF $\mathrm{Sp}(2,\mathbb{F}_2)$ WITH THEIR ACTION ON $V_{\text{gauge}} = \mathrm{span}\{v_X, v_Z\}$ AND CIRCUIT INTERPRETATION FOR THE $[[4,1,1,2]]$ BACON-SHOR CODE.

N	Action on (v_X, v_Z)	Circuit on gauge qubit
$\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$v_X \to v_X,\ v_Z \to v_Z$	Identity
$\begin{bmatrix}1&1\\0&1\end{bmatrix}$	$v_X \to v_X + v_Z,\ v_Z \to v_Z$	Phase (S) gate
$\begin{bmatrix}1&0\\1&1\end{bmatrix}$	$v_X \to v_X,\ v_Z \to v_X + v_Z$	Dual phase
$\begin{bmatrix}0&1\\1&0\end{bmatrix}$	$v_X \to v_Z,\ v_Z \to v_X$	Hadamard (H)
$\begin{bmatrix}1&1\\1&0\end{bmatrix}$	$v_X \to v_X + v_Z,\ v_Z \to v_X$	HS
$\begin{bmatrix}0&1\\1&1\end{bmatrix}$	$v_X \to v_Z,\ v_Z \to v_X + v_Z$	SH

C. Counting Solutions

Applying Theorem 4 with $r = 2$, $g = 1$:

$$|\mathrm{Sol}(U_L)| = 2^{r(r+1)/2} \cdot |\mathrm{Sp}(2,\mathbb{F}_2)| = 2^3 \cdot 6 = 48. \tag{48}$$

We contrast this with two incorrect applications of the stabilizer LCS theorem:

1) **Naïve stabilizer count** (treating $W_{\mathcal{G}}$ as isotropic with $r' = m - k = 3$): the LCS theorem does *not* apply because $W_{\mathcal{G}}$ is not isotropic (Remark 1). The formula $2^{r'(r'+1)/2}$ with $r' = 3$ gives $2^6 = 64$; this number is meaningless in the subsystem context because it applies to a different (hypothetical isotropic) code.
2) **Corrected stabilizer count** (applying the LCS theorem only to the isotropic stabilizer subspace $W_{\mathcal{S}}$ with $r = 2$): $2^{r(r+1)/2} = 2^3 = 8$. This is a valid group-theoretic count, but it *undercounts* the true solution space by the gauge factor $|\mathrm{Sp}(2,\mathbb{F}_2)| = 6$, since it ignores all circuits that permute the gauge generators non-trivially.

Our formula recovers the exact count **48**.

D. The Gauge Orbit: $\mathrm{Sp}(2,\mathbb{F}_2) \cong S_3$

$\mathrm{Sp}(2,\mathbb{F}_2) = \mathrm{SL}(2,\mathbb{F}_2)$ has order 6. The identification $\mathrm{SL}(2,\mathbb{F}_2) \cong S_3$ follows by noting that $\mathrm{SL}(2,\mathbb{F}_2)$ acts faithfully on the three non-zero vectors of $\mathbb{F}_2^2$ by left multiplication, giving an injection into S_3; equality of orders gives the isomorphism. The six elements, acting on the basis $\{v_X, v_Z\}$ of V_{gauge}, are listed in Table II. The lift $\sigma(N)$ acts as N on $\{v_X, v_Z\}$ and as the identity on all other basis vectors.

The Hadamard gauge element ($N = \left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$) exchanges X- and Z-type gauge generators: $v_X \mapsto v_Z$, i.e., $g_{X,1} \leftrightarrow g_{Z,1}$. This produces a physically distinct family of circuits that cannot be discovered within the stabilizer-only LCS framework.

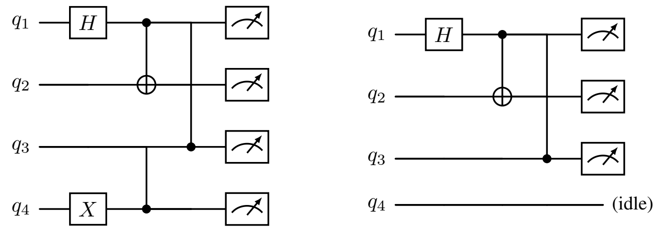

Fig. 2. Two physical realizations of the syndrome-extraction circuit for the logical $\bar{Z}_L$ Pauli on the $[[4,1,1,2]]$ Bacon–Shor code (measurements at the end extract the stabilizer eigenvalue). **Left** (stabilizer-sector solution): qubit q_4 participates via the gauge generator $g_{Z,2} = Z_2Z_4$, appearing as a two-qubit gate. **Right** (gauge-orbit solution via $\sigma(N)$ with $N = \left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$): the gauge transformation swaps $v_Z \leftrightarrow v_X$, reorganizing syndrome extraction using only $g_{Z,1} = Z_1Z_3$ and $\bar{Z}_L = Z_1Z_2$. Qubit q_4 is entirely idle and omitted from the compiled circuit, benefiting hardware with a degraded fourth qubit.

E. Compilation Advantage: Avoiding a Faulty Qubit

Consider synthesizing the unitary implementing the logical $\bar{Z}_L$ Pauli operator on the code space, when physical qubit 4 has degraded coherence. We ask: which elements of $\mathrm{Sol}(\bar{Z}_L)$ avoid two-qubit gates on qubit 4?

Among the 8 stabilizer-sector solutions (elements of $M_0 \cdot H_{\text{stab}}$), all require the gauge generator $g_{Z,2} = Z_2Z_4$ or $g_{X,2} = X_3X_4$ as building blocks, since the stabilizer $S_Z = g_{Z,1}g_{Z,2}$ involves qubit 4 and the stabilizer-only framework cannot reorganize its internal factors. Thus *every* stabilizer-sector solution uses qubit 4.

Among the gauge-orbit solutions with $N = \left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$, the gauge transformation swaps $v_Z \leftrightarrow v_X$, reorganizing syndrome extraction onto qubits $\{1,2,3\}$ only and leaving qubit 4 entirely idle (Fig. 2). This solution is inaccessible to the stabilizer-only LCS framework, and concretely illustrates the compilation advantage that gauge degrees of freedom provide.

F. Explicit 8×8 Symplectic Matrices

For completeness, we exhibit the 8×8 binary symplectic matrix $\sigma(H_{\text{gauge}})$ corresponding to the Hadamard gauge element $N = \left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$. In the ordered basis $(v_X, v_Z, s_X, s_Z, \bar{X}_L, \bar{Z}_L, r_X, r_Z)$:

$$\sigma(H_{\text{gauge}}) = \begin{pmatrix} \begin{matrix}0&1\\1&0\end{matrix} & 0 \\ 0 & I_6 \end{pmatrix}. \tag{49}$$

This is block-diagonal with the swap $\left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$ in the gauge block and identity elsewhere. One verifies $\sigma(H_{\text{gauge}})\,\Omega\,\sigma(H_{\text{gauge}})^T = \Omega$ directly, confirming membership in $\mathrm{Sp}(8,\mathbb{F}_2)$. Multiplied against any of the 8 stabilizer-sector solutions M_i, it yields 8 distinct gauge-orbit physical circuits.

VI. DISCUSSION AND CONCLUSION

A. Summary

We have extended the Logical Clifford Synthesis framework to $[[m,k,g,d]]$ subsystem stabilizer codes. The central

TABLE III
SOLUTION COUNTS FOR REPRESENTATIVE SUBSYSTEM CODES. $r = m - k - g$; STABILIZER FACTOR $= 2^{r(r+1)/2}$; GAUGE FACTOR $= |\mathrm{Sp}(2g, \mathbb{F}_2)|$.

Code	m	k	g	Stab. factor	Gauge factor
$[[4, 1, 1, 2]]$ Bacon-Shor	4	1	1	8	6
$[[9, 1, 4, 3]]$ Bacon-Shor	9	1	4	1,024	$\approx 4.7\times 10^{10}$
$[[16, 1, 9, 4]]$ Bacon-Shor	16	1	9	$\approx 3.4\times 10^6$	$\approx 1.2\times 10^{44}$
Honeycomb (small) [7]	12	2	4	32,768	$\approx 4.7\times 10^{10}$

result is Theorem 4: the group H_{gauge} of physical symplectic realizations of any logical Clifford decomposes as $H_{\mathrm{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2)$, yielding the closed-form solution count $2^{r(r+1)/2} \cdot |\mathrm{Sp}(2g, \mathbb{F}_2)|$. The proof rests on (i) the symplectic decomposition $W_{\mathcal{G}} = W_{\mathcal{S}} \oplus V_{\mathrm{gauge}}$ (Lemma 3), (ii) the construction of an explicit splitting section $\sigma : \mathrm{Sp}(2g, \mathbb{F}_2) \to H_{\mathrm{gauge}}$, and (iii) the centrality of H_{stab} in H_{gauge} (which upgrades the semidirect to a direct product).

B. Implications for Compilation

Table III shows solution counts for several subsystem codes of practical interest. The gauge factor $|\mathrm{Sp}(2g, \mathbb{F}_2)|$ grows as $\Theta(2^{g^2})$, vastly exceeding the stabilizer factor $2^{r(r+1)/2}$ for large g. For the $[[9, 1, 4, 3]]$ Bacon-Shor code, the gauge factor is $\approx 4.7 \times 10^{10}$, giving enormous additional compilation freedom compared to the stabilizer-only count of 1024.

The separability result (Corollary 6) has direct algorithmic value: for hardware-specific cost functions that decompose additively over stabilizer and gauge subcircuits, joint optimization reduces to two independent subproblems. For codes like the Bacon-Shor family, the gauge subcircuit acts on a $2g$-dimensional space and can be optimized with algorithms tailored to $\mathrm{Sp}(2g, \mathbb{F}_2)$ [16], [17].

C. Limitations and Open Problems

(i) Non-Abelian gauge groups with restricted gauge factors. Remark 5 noted that for Floquet codes [7], the image of φ may be a proper subgroup of $\mathrm{Sp}(2g, \mathbb{F}_2)$. Characterizing this image for specific Floquet codes, and determining whether the product structure $H_{\mathrm{gauge}} \cong H_{\mathrm{stab}} \times \mathrm{Im}(\varphi)$ still holds in those cases, is an important open problem.

(ii) Code design for compilation diversity. Theorem 4 introduces a new code design criterion: which $[[m, k, g, d]]$ codes with a given distance d maximize the gauge factor $|\mathrm{Sp}(2g, \mathbb{F}_2)|$? Since $|\mathrm{Sp}(2g, \mathbb{F}_2)| = 2^{g^2} \prod_{i=1}^{g}(4^i - 1)$ is strictly increasing in g, codes with larger gauge qubit count (at fixed m, k, d) offer more compilation freedom. The interplay between g and d (larger g sometimes forces lower d) defines a new trade-off space.

(iii) Non-Clifford synthesis. Extending the present framework to the third level $\mathcal{C}^{(3)}$ of the Clifford hierarchy (CCZ, T, CS gates) requires replacing the symplectic group with the orthogonal group $O(2m, \mathbb{F}_2)$ [3], [22]. Deriving the analogue of Theorem 4 for $\mathcal{C}^{(3)}$ on subsystem codes would have direct implications for magic-state distillation protocols.

(iv) Noise-adaptive heuristics. The full $|\mathrm{Sol}(U_L)|$ solution set is too large to enumerate exhaustively for large codes. Practical compilation will require heuristic search within the coset $M_0 \cdot H_{\mathrm{gauge}}$, exploiting the product structure to search the stabilizer and gauge factors independently. Developing such heuristics, analogous to the noise-adaptive mappings of [2], [9], is a natural next step. Recent work on QLDPC subsystem codes [23] demonstrates that full Clifford groups can be compiled transversally with $O(m)$ syndrome rounds, a setting where our gauge-orbit enumeration provides the complete solution space for each compiled gate.

(v) Software implementation. The open-source LCS implementation of [3] can be extended by adding Phase 2 of Algorithm 1: a symplectic Gram-Schmidt routine within $W_{\mathcal{G}}/W_{\mathcal{S}}$, and an enumerator for $\mathrm{Sp}(2g, \mathbb{F}_2)$ via its transvection generators.

D. Conclusion

The direct product structure $H_{\mathrm{gauge}} \cong H_{\mathrm{stab}} \times \mathrm{Sp}(2g, \mathbb{F}_2)$ is the central insight of this paper. It shows that subsystem codes carry strictly more implementation diversity than their stabilizer skeletons—by an exact factor of $|\mathrm{Sp}(2g, \mathbb{F}_2)|$—that this extra diversity is algebraically clean (a symplectic group), and that it can be exploited independently of the stabilizer degrees of freedom. The $[[4, 1, 1, 2]]$ Bacon-Shor example demonstrates concretely that gauge-orbit solutions can enable hardware-efficient implementations entirely inaccessible to the stabilizer-only framework. As subsystem codes take on a larger role in near-term and fault-tolerant hardware, systematic tools for exploiting their full implementation freedom will be increasingly valuable.

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